



Components in a smart grid; Concept (Courtesy: National Instruments)

sors are used to collect information on the performance and health of station equipment including transformers and circuit breakers.

Companies like TI, ADI and Intel offer a range of energy metering ICs, power-line communication systems, data concentrators, microcontrollers for home-area networking, in-home display systems, smart meters, IoT gateways and so on.

TI offers various ARM-based system-on-chip and system-on-module solutions and software for the smart grid, as well as PLC and data concentrators. Intel offers a series of chips with remote management capabilities. Their active management technology implements a special circuit in Intel chipsets, which enables the system to be remotely accessible even when it is powered off or the software is corrupt, using an out-of-band link. The cross-platform solution can apparently be used with energy generation equipment, control systems and home energy gateways. A range of Intel Xeon, Intel Core and Intel Atom processors are used across grid infrastructure.

Field-programmable gate arrays (FPGAs) are also used widely in smart grid implementations. They offer much-needed flexibility and integration in a diverse system, and in cases

such as NI's solution, they can be customised as well as upgraded in the field, without changing the hardware, to behave as a number of different 'personalities,' such as PMU, power quality, smart switch, recloser, etc.

Test and Measurement. The sensing aspects of a smart grid are basically the preserve of the T&M world, which constantly improves capabilities with better embedded instrumentation systems. "Developments in the T&M sector have enabled better sensing—speed, accuracy, synchronisation across long distances; better communication—high-throughput, reliable long-distance data transfer and field upgradeability," says Mohanram.

Communications. The ability of the nodes to communicate and coordinate with each other and a central system makes a grid smart, or otherwise. Both wired power line (PLC) and wireless (Wi-Fi, ZigBee and Sub-1GHz) communication are used in a smart grid. The 2.4GHz IEEE 802.15.4 standard is quite popular for wireless communications in smart grids, due to its low data rate but extremely long battery life and very low complexity. Several network protocols including ZigBee, 6LoWPAN, WirelessHART and ISA100.11a run on 802.15.4-based networks. Thakurdesai shares that, "In

smart substations, the new standard for communication is IEC 61850 and for PLC, standards like Prime and G3, which are certified with TI products."

Smart meters. Smart meters can be seen as the face of the smart grid, what consumers see. These intelligent bi-directional devices help both the consumer and the utility. The former can understand their consumption patterns and alter it effectively, so as to conserve power and capitalise on utility-given discounts. Utilities, on the other hand, can use it to effectively plan their distribution, control power consumption, understand consumers, avoid power theft, implement variable pricing and so on.

And big data too...

"As the smart grid evolves with sensors distributed across the grid collecting huge amounts of data, drawing accurate and meaningful conclusions from such a large amount of data is an interesting problem, and the term Big Data describes this phenomenon," says Mohanram. The smart grid too crunches huge amounts of data coming from analogue devices. "We call it Big Analog Data solutions, described using a three-tier solution architecture. The first two tiers are what inherit from the measurement world, namely, the sensors and the system nodes. Once through tier 2, the data moves past 'the edge' and usually hits a network switch and then a server where it can be stored and undergoes further analysis. This is where customers realise great value from the data they've acquired, and where many engineering, scientific and business decisions are made. This tier also includes the Cloud, a growing and appealing IT infrastructure that powers the smart grid," explains Mohanram.

The data captured from the grid can be analysed and put to use at various fronts:

1. Analysis of AMI data with a customer-focus can help in theft detection and revenue protection, demand-response management, happy-hour pricing and other variable pricing schemes to